# Tutorial 03 — Instructor notes

ENVX2001 – Applied Statistical Methods

Semester 1

# Contents



Companion document for tutors running Tutorial 03. Mirrors each exercise with talking points, expected output at each checkpoint, and common problems.

Student document: Tutorial 03

# Before you start

Get these sorted before students arrive:

- Have RStudio open with a fresh session.
- No packages need installing for this tutorial. The `sleep` dataset is built into R.
- Run `str(sleep)` and make sure you can see the 20-row structure (10 patients measured under two treatments, 20 observations). You will use this dataset throughout Exercise 1.
- Skim the document if you have not… Exercise 1 covers assumption checking, Exercise 2 covers ANOVA table interpretation. They are independent, so if you run short on time you can abbreviate one without losing the other.

# Exercise 1: Checking assumptions in R

## Diagnostics code

### What to do

Walk through both sets of code (t-test and ANOVA) live. Have students run each block and look at the output before you explain it.

**t-test code:** Start with `str(sleep)` so everyone sees the data structure. Then subset into two groups and produce QQ plots and Shapiro-Wilk tests for each.

CODE

```r
str(sleep)

group1 <- subset(sleep, group == "1")
group2 <- subset(sleep, group == "2")

qqnorm(group1$extra, main = "Group 1")
qqline(group1$extra, col = "red")

qqnorm(group2$extra, main = "Group 2")
qqline(group2$extra, col = "red")

shapiro.test(group1$extra)
shapiro.test(group2$extra)
```

**ANOVA code:** Fit the model, then check residuals directly.

CODE

```r
model <- aov(extra ~ group, data = sleep)

qqnorm(residuals(model), main = "")
```

```r
qqline(residuals(model), col = "red")
shapiro.test(residuals(model))
```

**Equal variance:** Boxplot and SD ratio.

CODE

```r
boxplot(extra ~ group, data = sleep)

group1 <- subset(sleep, group == "1")
group2 <- subset(sleep, group == "2")
max(sd(group1$extra), sd(group2$extra)) /
  min(sd(group1$extra), sd(group2$extra))
```

### What to say

- **Normality (QQ plot):** Points should fall roughly along the red line. With only 10 observations per group, some scatter is expected. The key is whether there is a systematic curve (indicating skew) or extreme outliers pulling away from the line.
- **Normality (Shapiro-Wilk):** Both p-values are well above 0.05 (around 0.4 and 0.3), so we do not reject normality. Emphasise that this test has low power with small samples, so a non-significant result does not prove normality — it just means we have no evidence against it.
- **Equal variance:** The SD ratio for the sleep data is about 1.5, which is below the rule-of-thumb threshold of 2. The boxplots show similar spread. This assumption is reasonable.
- **Independence:** Mention that this cannot be checked from data. It depends on the study design. For the sleep data, each patient received both drugs (paired design), which actually violates the independence assumption for a standard two-sample t-test. You do not need to go deep on this, but if a sharp student asks, it is a good teaching moment.
- **t-test vs ANOVA approach:** Point out that checking residuals from `aov()` is simpler than subsetting groups manually. Both approaches give the same conclusion. In the lab, students will use the ANOVA approach exclusively.

### Expected output

- QQ plots: points roughly following the line, with minor deviations at the tails.
- Shapiro-Wilk: p-values > 0.05 for both groups and for the ANOVA residuals.
- SD ratio: approximately 1.5.

### Common issues

> ⚠ Warning
>
> **"What does the red line mean?"** The red line is the theoretical quantile line. If the data were perfectly normal, all points would sit on this line. Deviations indicate departures from normality. Curvature suggests skew; S-shapes suggest heavy or light tails.

⚠ Warning

**Students confuse the Shapiro-Wilk null hypothesis.** The null is that the data ARE normal. A small p-value means the data are NOT normal. This is the opposite of what students expect (they are used to small p-values being "significant" in a positive sense). Spell this out explicitly.

## Reading diagnostics (widget)

### What to do

Let students work through 5 to 10 examples from the interactive widget. Circulate and check that they are getting the right answers. If the group is comfortable, speed through this section.

### What to say

- The widget shows QQ plots, boxplots, or SD ratios for randomly generated data with 2 to 4 groups.
- For QQ plots: look for systematic curvature, not individual point scatter. Small samples will always have some wobble.
- For boxplots: compare the box heights (IQR). If one group has a much larger box, equal variance may be violated.
- For SD ratios: below 2 is the rough threshold.
- Remind students that these are guidelines, not hard rules. With small samples, violations are harder to detect and the tests are more robust to minor departures anyway.

### Common issues

⚠ Warning

**Widget does not load.** The widget requires JavaScript. If a student's browser blocks scripts, it will not render. They can follow along with a neighbour or skip ahead to Exercise 2.

# Exercise 2: The F ratio and degrees of freedom

## Maize and wheat ANOVA tables

### What to do

Walk through the two pre-computed ANOVA tables (maize planting density, wheat fertiliser blends). Students do not run any code here — they read and interpret.

### What to say

- **F ratio as signal-to-noise:** $F = MS_{trt}/MS_{res}$. This is the single most important concept in the exercise. The treatment mean square measures how different the group means are. The residual mean square measures the random scatter within groups. F tells us whether the signal (group differences) is large relative to the noise (within-group scatter).
- **Maize table:** $F = 152.96/4.97 = 30.77$. The between-group signal is 30 times larger than the noise. P < 0.001. Clear evidence that planting density affects yield.
- **Wheat table:** $F = 5.70/5.00 = 1.14$. The signal is about the same size as the noise. P = 0.35. No evidence that fertiliser blend matters.
- **Contrast the two:** Same design (3 groups, 5 reps each), same residual df (12), but completely different conclusions. The ANOVA table makes the difference obvious.
- **Degrees of freedom:** Treatment df = groups minus 1. Residual df = total observations minus groups. From these two numbers alone, you can reconstruct the experimental design.
- **Sum of squares partition:** Treatment SS + Residual SS = Total SS. All variation in the data is split into explained (by the grouping) and unexplained (random noise). This is worth mentioning but do not dwell on it — students will see it again in later weeks.

### Expected understanding

Students should be able to:

- Calculate F from the mean squares.
- State whether the result is significant from the p-value.
- Work out the number of groups and total observations from the degrees of freedom.

## Try it yourself (tomato watering regimes)

### What to do

Give students 3 to 5 minutes to work through the three questions independently, then go through the answers as a group.

### Answers

1. Treatment df = 4 − 1 = 3. Residual df = 24 − 4 = 20. (4 groups, 6 plants each = 24 total.)
2. $F = 45.2/8.1 = 5.58$.
3. $F_{obs} = 5.58 > F_{crit} = 3.10$, so we reject $H_0$ at $\alpha = 0.05$. The watering regimes have a significant effect on tomato yield.

### What to say

- Walk through the df calculation first. Students often forget that total N = groups × replicates per group.
- For the F ratio, emphasise that it is just division. Nothing fancy.
- For significance: comparing F to the critical value is the old-fashioned way (before computers gave us p-values). It is worth knowing because it builds intuition about what F "means" — how many times larger is the signal than the noise?
- If time allows, show the R command: `qf(0.05, 3, 20, lower.tail = FALSE)` gives 3.10. This connects the abstract critical value to something they can compute.

### Common issues

> ⚠ Warning
>
> **Students compute total df instead of residual df.** They may say residual df = 24 − 1 = 23 (total df). Remind them that residual df = N − t, where t is the number of groups. Treatment df + Residual df = Total df.

> ⚠ Warning
>
> **"Is 5.58 a big F?"** It depends on the degrees of freedom. F = 5.58 with df = (3, 20) is significant at 0.05. The same F with df = (3, 5) might not be. This is why we need the critical value or p-value rather than just eyeballing F.

# Putting it all together

## What to say

- Two big ideas from today: (1) we check assumptions with QQ plots, Shapiro-Wilk, boxplots, and SD ratios; (2) we read an ANOVA table by computing F and extracting the design from the degrees of freedom.

- In the lab this week, students will fit ANOVA models in R, check assumptions on real data, and use `emmeans` for post-hoc comparisons. The tutorial has given them the conceptual tools; the lab gives them the hands-on practice.
- If students found the F ratio and degrees of freedom calculations straightforward, reassure them that these are examinable and worth practising. If they struggled, point them to the lecture slides (Lecture 03b) which cover the same material with additional worked examples.